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The Pictou Red Beds' Pennsylvanian Pole: Could Phanerozoic Rocks in the Interior United States Be Remagnetized?

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The Pennsylvanian red beds of the Pictou Group on Prince Edward Island, Canada, retain a stable remanence in hematite that gives a paleomagnetic pole position at 42°N, 131°E (dp=2°, dm=4°). Positive conglomerate and contact tests along with dual polarities show that the remanence is primary and of Stephanian age (286-306 Ma). For elongated pole distributions through time, Bingham statistics is more suitable than Fisher statistics. Bingham statistics shows that there is a consistent discordance in the Pennsylvanian to Late Triassic poles from the Appalachian orogenic belt and interior craton of North America. This discordance is most readily explained by younger partial remagnetization, or possibly complete remagnetization, in many rock units, especially red beds of the interior craton. Bingham analysis also shows that there is a significant difference between the Colorado Plateau and interior craton that is most readily explained by a Laramide 5.5° ± 2.9° clockwise rotation of the plateau relative to the craton.

INTRODUCTION

The Kiaman (Late Carboniferous-Permian) apparent polar wander (APW) path for North America is defined by relatively numerous paleopoles [*Irving and Irving*, 1982; *Van der Voo*, 1990]. As new poles add to the path's precision, it becomes possible to detect significant differences between the poles from the different terranes to reveal major tectonic events. Estimation of the Colorado Plateau rotation within the North America craton is one such example [*Steiner*, 1986, 1988; *Bryan and Gordon*, 1986, 1990].

We sampled the lowermost Pictou red beds with a well-defined mid Pennsylvanian (Stephanian) age on Prince Edward Island (PEI) to add to the Kiaman APW path for the craton. These red beds have been studied paleomagnetically by *Black* [1964] and *Roy* [1966]. Lacking stratigraphical control, they sampled the entire mid-Pennsylvanian to mid-Early Permian sequence. Also they used alternating field (AF) and low-temperature thermal demagnetization that are unsuitable for red bed studies by current standards [*Roy and Morris*, 1983]. Recently, *Symons* [1990] in a paleomagnetic report of the uppermost Pictou red beds found the Early Permian poles from the northeastern and southwestern North American craton to be significantly different and suggested the possibility of relative motion between the two parts of the craton. This study of the lowermost Pennsylvanian Pictou red beds extends that study.

GEOLOGY

The Pictou Group on PEI consists of flat-lying red beds that *van der Poll* [1983] subdivided into four upward fining sequences and designated as megacycles I to IV. Each megacycle is 300-400 m thick with a basal pebble conglomerate with clasts consisting of quartzite, rhyolite and other rock types. We sampled only megacycle I red beds except for the overlying basal conglomerate and some tilted red beds of lowermost megacycle II (Figure 1). The paleontological and palynological evidence defines a mid-Pennsylvanian (Stephanian, 286-306 Ma) age for megacycle I [*Palmer*, 1983]. Their poor induration and gentle dip (3°-4°) indicate that both metamorphism and structural deformation are negligible [*van der Poll*, 1983].

SAMPLING

Twenty sites, each represented by four oriented blocks, were sampled in megacycle I red beds around the southwestern shore of the PEI (Figure 1). The shoreline sections are typically 2-10 m thick, and the blocks were taken randomly from different layers in each bluff. Detailed lithologic correlation between sites is not possible. At sites 6 and 7, 19 oriented walnut-sized clasts were collected from basal megacycle II for a conglomerate test. Three more sites of six or seven blocks each were collected from megacycle II red beds with dips of 7°-8° for a fold test. In most cases, four 2.5-cm specimens were drilled from each red bed block and one or two 0.9-cm specimens were drilled from each clast. A coat of sodium metasilicate was occasionally used to prevent disintegration during drilling [*Symons et al.*, 1989].

EXPERIMENTAL METHODS

Natural remanent magnetizations (NRM) were measured on an automated CTF cryogenic magnetometer. Thermal demagnetization was done using a Schonstedt TSD-1 unit, and AF demagnetization using a Sapphire Instruments SI-3 unit. Isothermal remanent magnetizations (IRM) were induced in fields up to 900 mT using a Sapphire Instruments SI-5 pulse magnetizer. Remanent magnetization directions were determined by principal component analysis [*Kirschvink*, 1980], requiring a specimen's maximum angular deviation to be <10°. Mean directions and poles were calculated using standard *Fisher* [1953] statistics.

RESULTS

Thermal Step Demagnetization

All red bed specimens were thermally demagnetized. Initially, two specimens from each site were demagnetized in 11 steps up to 680°C with most steps above 400°C. The remaining specimens were

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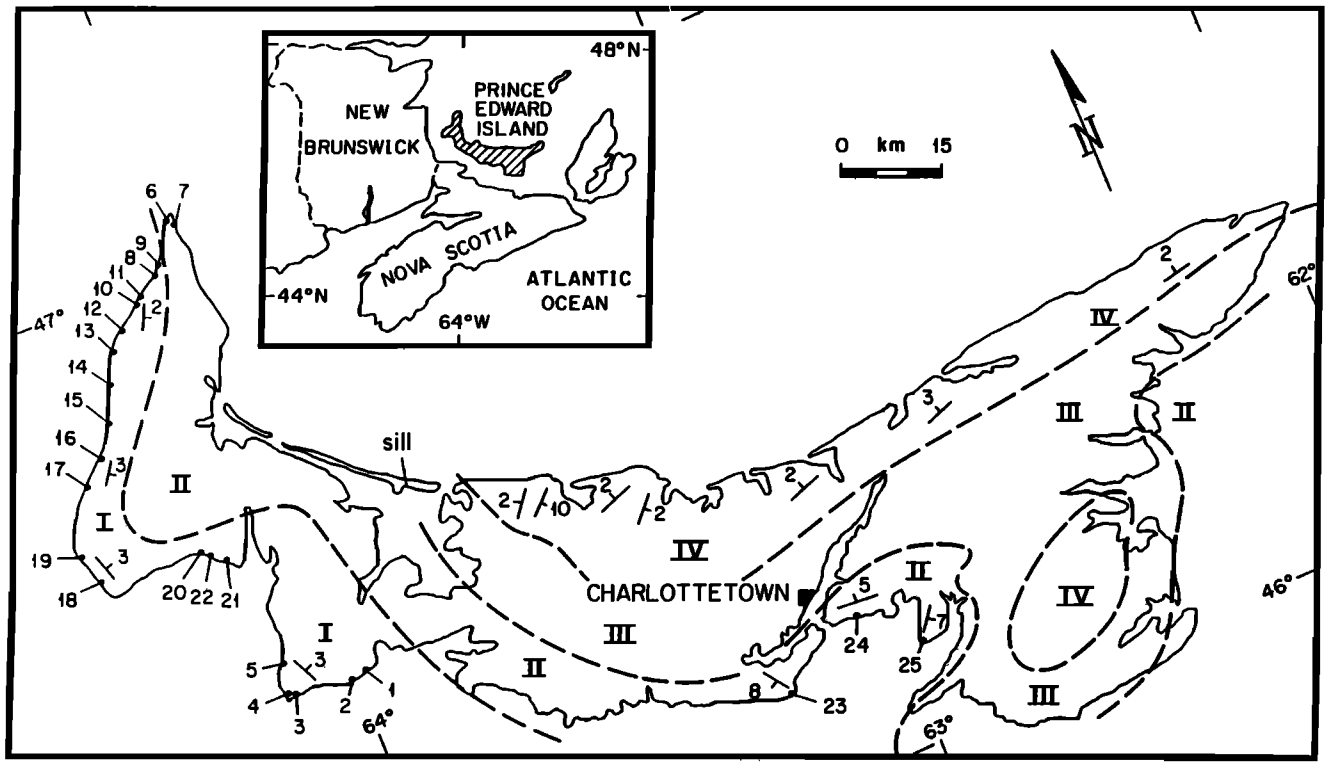


Fig. 1. Map of geology of Prince Edward Island with sampling locations. I,II,III, and IV denote the megacycles in the Pictou red Beds [after *van de Poll*, 1983].

demagnetized at 550, 615, 640, and 665°C. Almost all specimens behaved very similarly at higher temperatures. For most specimens, first a steep or intermediate inclined normal remanence component was removed with blocking temperatures up to about 450-550°C, and then a characteristic component with a shallow southerly inclination was isolated and removed by temperatures up to about 670°C (Figure 2). These characteristic directions were averaged to obtain the site mean directions (Table 1 and Figure 3*b*). For sites 5, 13 and 21, the individual specimens gave well-defined characteristic directions, but they were dispersed to give large radii of 95% confidence (α_{95}) about the site means, and so these means were excluded from the unit mean calculation. No characteristic directions were isolated from site 24.

Remanence Polarity

Almost all red bed specimens gave a reversed remanent magnetization polarity except for one block with a normal polarity from each of sites 14 and 17. After reversal the two normal blocks have a mean declination (D) of 168° and inclination (I) of 5° which is the same as the unit mean direction. With so few normal polarity specimens, it is inappropriate to use the statistical test of *McFadden and Jones* [1981] to confirm a reversal test.

Isothermal Remanent Magnetization

The IRM acquisition curves for 10 representative specimens are typical of hematite (Figure 4), confirming the thermal demagnetization results. AF step demagnetization of the IRM to 50 mT further indicates that the remanence carrier in the red beds is dominantly fine grained hematite.

Conglomerate and Fold Tests

Specimens from the 19 clasts were tested using both thermal and AF step demagnetization, depending on the rock type (Figure 2*c*).

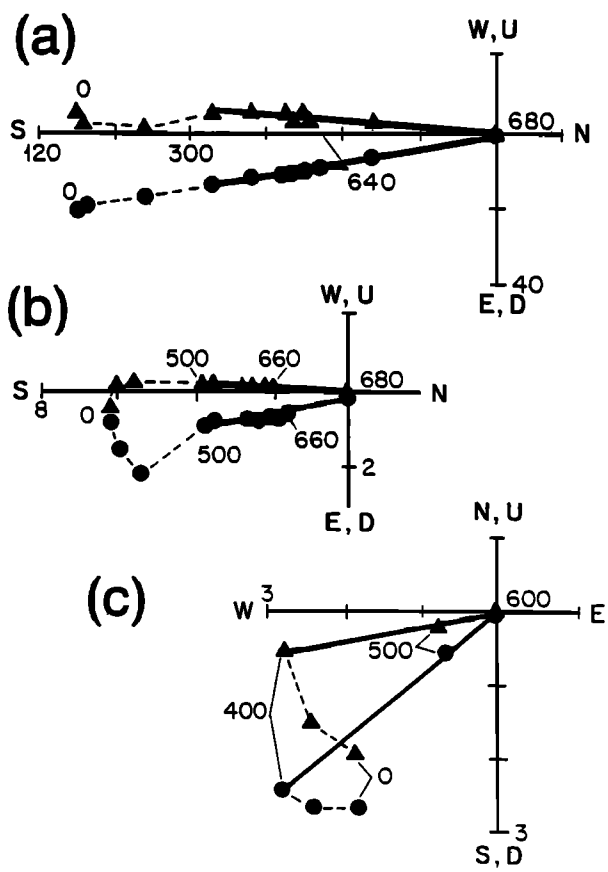


Fig. 2. Orthogonal vector demagnetization diagrams for example specimens from the red bed (*a*) sites 8 and (*b*) 12 and (*c*) from a conglomerate clast from site 6. Circles and triangles represent horizontal and vertical projections, respectively. The axial units are in 10^{-2} A/m.

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TABLE 1. Site Mean Remanence Directions After Tilt Corrections

Site	N	D/I, deg	α_{95}, deg	κ
1	7	168/-08	16	15
2	10	173/-04	14	13
3	12	162/+11	4	132
4	11	160/+07	5	95
5	7	172/-01	27	8
6	10	198/+27	72	2
7	11	282/+44	55	2
8	11	175/+05	6	58
9	6	174/+02	12	34
10	11	170/-06	5	85
11	12	171/-09	10	18
12	8	166/-07	7	63
13	4	175/+01	38	7
14	12	164/+00	5	89
15	11	177/-09	11	18
16	9	167/+02	16	12
17	9	172/-03	14	14
18	12	160/-01	8	34
19	12	168/+04	8	32
20	8	168/+08	8	55
21	12	135/+64	34	3
22	8	167/+06	4	165
23	16	176/+03	7	29
25	16	163/+13	8	21

Unit mean is D = 169°, I = 0°, α_{95} = 4°, N = 17, and pole position is 42°N, 131°E, dp = 2°, dm = 4°. N is the number of specimens; the unit mean excludes date from dites 5, 6, 7, 13, 21, 23, 24, and 25 as explained in the text.

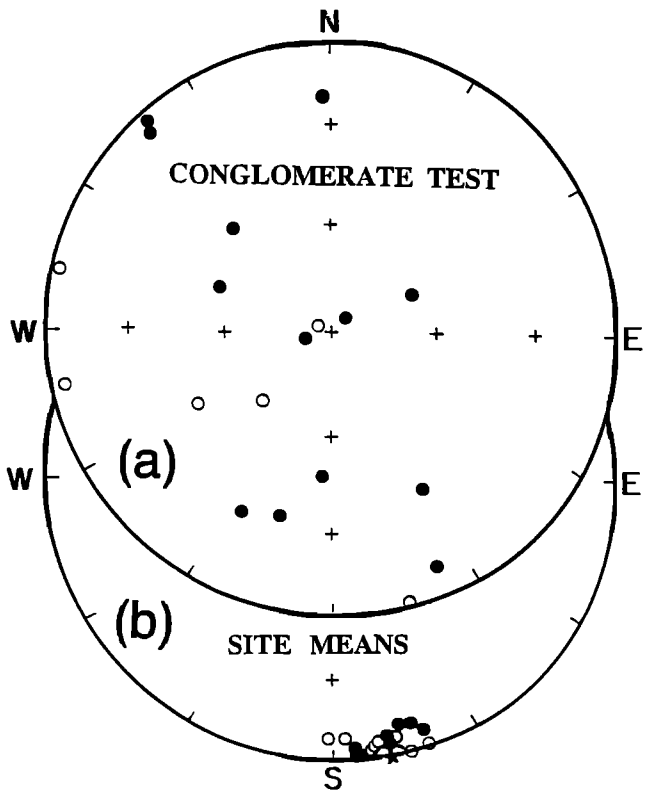


Fig. 3. Equal-area stereograms of the (*a*) mean directions for 19 conglomerate clasts; and (*b*) site means for 17 accepted red bed sites with the unit mean (star) and its cone of 95% confidence. Open and solid symbols denote projections on the upper and lower hemisphere, respectively.

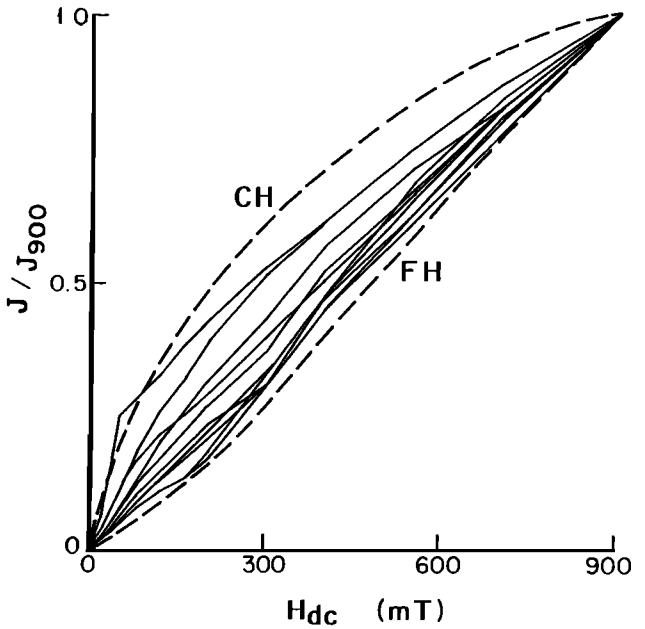


Fig. 4. Isothermal remanent magnetization (IRM) showing acquisition of IRM up to 900 mT. Type curves are for coarse-grained (CH) and fine-grained (FH) hematite.

Most specimens contain considerable hematite alteration, presumably acquired during transport prior to deposition. The characteristic directions for the clasts are randomly directed with >99% confidence (Figure 3*a*), indicating that the clasts have not been remagnetized since deposition. Thus the conglomerate test is positive [*Watson*, 1956]. A fold test was attempted using the characteristic directions from sites 23 and 25. However tilt correction does not significantly change the precision parameter [*McFadden and Jones*, 1981], and so the test is inconclusive although their mean directions are rotated closer to the unit mean direction after correction.

DISCUSSION AND TECTONIC IMPLICATIONS

The Pictou Pole

The 17 accepted site mean remanence directions from megacycle I red beds give a mean of D = 169°, I = 0° (α_{95} = 4°) after tilt correction (Table 1). Three observations indicate that it is a primary Stephanian magnetization. First, the positive conglomerate test points to a pre-megacycle II age, defined by paleontology and palynology as earliest Early Permian [*Barss et al.*, 1963; *Hacquebard*, 1972]. Second, *Symons* [1990] has found a positive contact test for the overlying megacycle IV red beds. The mean direction for the megacycle I red beds is also significantly different from that for the Malpeque Bay sill and baked zone [*Larochelle*, 1967; *Symons*, 1990], indicating that their remanence also predates the Late Permian age (247±1 Ma) of the sill. Third, the occurrence of antiparallel normal polarity directions, although present in only two blocks, suggests, if not strongly, that the unit mean direction is primary.

The unit mean direction for the megacycle I red beds yields a pole position of 42°N, 131°E (dp=2°, dm=4°) (latitude, longitude (radii of oval of 95% confidence))(Table 1). *Roy* [1966] found a statistically similar pole of 42°N, 133°E (dp=3°, dm=6°) for the Pictou Group overall but with a larger oval of confidence. However, his six megacycle I sites (A2-A7) alone give a pole at 41°N, 141°E (dp=4°, dm=8°) that is significantly different from our pole. This difference

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TABLE 2. Paleomagnetic Poles Used in This Study

Rock Unit	Pole °E/°N	Region§	Rock Unit	Pole °E/°N	Region§
Middle-Upper Triassic			*Lower Permian*		
Newark Basin, lower red beds	102/54	AP	Upper Pictou red beds[4]	127/42	AP
Newark Basin (R-polarity)	089/49	AP	Cap aux Meules Formation[5]	129/45	AP
Abbott Pluton	092/48	AP	Middletown Pegmatites[6]	132/35	AP
Agamenticus Pluton	099/48	AP	Elephant Canyon Formation	120/44	CP
Chinle Formation, Arizona[1]	066/57	CP	Cutler Formation (Lisbon Valley)	118/44	CP
Chinle Formation, New Mexico[2]	087/60	CP	Halgaito Tongue[7]	117/49	CP
Chinle Formation, Utah	064/61	CP	Cutler Formation, Colorado	122/43	CP
Manicouagan Structure[3]	089/60	IC	Cutler Formation, Utah	116/44	CP
Chinle Formation, New Mexico	079/58	IC	Fountain and Lykins formations	119/48	CP
Popo Agie Formation	096/56	IC	Minturn and Maroon formations	115/43	CP
			Leonardian subset, Oklahoma only	119/52	IC
Lower Triassic			Casper Formation	123/51	IC
			Abo Formation	118/50	IC
Upper Maroon Formation	102/60	CP	Ingelside Formation	122/46	IC
Moenkopi Formation	102/60	CP			
Upper Moenkopi Formation	103/55	CP	*Upper Carboniferous*		
Upper Moenkopi drillcore	100/57	CP			
Moenkopi Formation, Arizona	101/58	CP	Morien Group	131/40	AP
Ankareh Formation	105/51	IC*	Cumberland Group	125/36	AP
Upper Red Peak Formation	105/49	IC*	Bonaventure Formation	133/38	AP
Lower Red Peak Formation	121/46	IC*	Hurley Creek Formation	125/39	AP
Chugwater Formation	114/47	IC*	Tormentine Formation	132/41	AP
Chugwater Formation	115/45	IC*	Piedmont Mafic Intrusions	121/39	AP
			Brush Creek Limestone	124/36	AP
Upper Permian			Churchland Pluton	126/34	AP
			Lower Pictou Group[8]	131/42	AP
Basic Sill, PEI	111/51	AP	Wescogame Formation (Supai)	119/47	AP
Hoskinnini Tongue	121/50	CP	Laborcita Formation	132/42	IC
Ochoan red beds	119/55	IC	Casper Formation	129/46	IC
Guadalupian red beds	125/51	IC	Mason Creek	131/37	IC
Dewey Lake Formation	126/51	IC			

References: 1, *Bazard and Butler* [1991]; 2, *Molina-Garza* et al. [1990]; 3, referenced by *Irving* and *Irving* [1982]; 4, *Symons* [1990]; 5, *Tanczyk* [1988]; 6, *de Boer and Brookings* [1972]; 7, *Farrell and May* [1969]; and 8, this study. Unreferenced poles are given by *Van der Voo* [1990].

§Region AP is for the Appalachian Orogen; CP is for the Colorado Plateau; and IC is for the interior craton of North America.

*These poles from the IC were not used for the Triassic comparison with the AP poles because there are not equivalent Lower Triassic AP poles.

likely stems from his use of 450°C thermal demagnetization because the characteristic remanence for most specimens is only well isolated above 500°C.

In a recent analysis, *Van der Voo* [1990] accepted 15 Pennsylvanian (Late Carboniferous) poles as reliable for APW path construction for the North American craton. Of these, the Dunkard Formation pole [*Helsley*, 1965] is deemed unreliable because most samples were not demagnetized and the Pictou red bed poles of *Roy* [1966] and *Black* [1964] are now superseded [*Symons*, 1990; this study]. Our lower Pictou red bed pole is similar to the other Pennsylvanian poles from the craton. However, the pole positions from the interior craton (IC) are noticeably different overall than the pole positions from the Appalachian orogenic belt (AP). This systematic difference leads us to analyze whether or not the pole distributions from the two segments of the craton are statistically significantly different and, if they are, what are the geologic implications.

Bingham Analysis of the Poles from the Craton

We used all available reliable poles from the Late Carboniferous to Late Triassic time span (Table 2). Our data base includes almost all the poles from the compilations of *Irving and Irving* [1982] and *Van der Voo* [1990], plus newly published poles but excluding poles from the Ouachita, Cordilleran, and Ellesmerian orogenic belts because of their uncertain tectonic correction. Also, previous geological and paleomagnetic studies have suggested that the Colorado Plateau is a rotated segment of the stable interior craton [*Hamilton*, 1981; *Steiner*, 1988; *Bryan and Gordon*, 1986, 1990]. Therefore, we divided the selected poles into three groups: the first is from the Appalachian orogenic belt and eastward (AP); the second is from the Colorado Plateau (CP); and the third is from elsewhere on the craton to the west of the Appalachians (IC) (Figure 5).

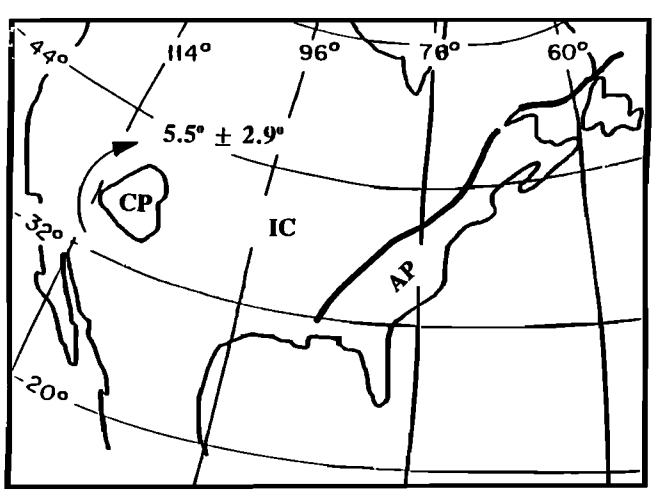


Fig. 5. Schematic of the Appalachian orogen (AP), the Colorado Plateau (CP), the interior craton (IC) of North America, and clockwise rotation of the Colorado Plateau.

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It is well accepted that the distribution of paleomagnetic poles representing a few million years or less can be described by *Fisher* [1953] statistics. However, if the age range of the poles is larger, then their distribution is no longer circular but becomes elongated along the trend of the APW path. Such an elongated distribution is more appropriately analyzed by Bingham statistics [*Bingham*, 1974; *Onstott*, 1980]. Therefore, we used Bingham statistics to compare the Late Carboniferous to Triassic poles for the IC and AP groups and the IC and CP groups. In Bingham statistics, the mean direction of the maximum concentration is the eigenvector direction corresponding to the maximum eigenvalue for the inertia of the moment and its confidence is represented by an oval about the mean. The results using Bingham statistics are listed in Table 3. Further, the difference between the mean poles with their ovals of confidence can be resolved to obtain the relative rotation and translation with the corresponding confidence intervals. The relative motions were calculated using a reference point at 46°N, 64°W for the AP and IC comparison, and at 38°N, 110°W for the CP and IC comparison (Table 4).

Pennsylvanian-Triassic APW path

One of the advantages of Bingham statistics over Fisher statistics is that it also gives the trend of the elongated pole distribution from the direction of the major axis of the confidence oval. This trend provides valuable information for the construction of the APW path and therefore on the direction of plate motion. The Pennsylvanian-Triassic segment of the North American APW path was treated as an approximately smooth track by *Gordon et al.* [1984] and *Bryan and Gordon* [1986, 1990], who assumed that there was no change in plate motion or equally in its Euler pole during the Pennsylvanian to Triassic time so that the APW path could be fitted by a small circle. From Bingham statistics, both the Pennsylvanian-Early Triassic and Middle-Late Triassic APW segments for the AP-IC and CP-IC comparisons generally follow the track indicated by the PEP small circle fitting, although the trend from the AP2 pole is slightly different which might attributed to the small number of poles available for this segment (Figures 6, 7*a* and 7*b*). Nevertheless, the fact that the four populations do not fit one track shows that treating the Pennsylvanian-Triassic segment of the path as a smooth track is only first-order approximation.

Appalachian Orogenic Belt Versus Interior Craton

Bingham statistical analysis indicates that the distributions of poles from the IC and AP for both the Pennsylvanian-Permian and Middle-Upper Triassic segments are significantly different at much greater than the 95% confidence level with the APW path for IC being parallel to but several degrees north of that of AP (Figure 7*a*). Assuming their characteristic remanent magnetization is primary then the apparent ages of the poles from the AP for the Pennsylvanian-Permian rock units are about 16 m.y. older on average than those from the IC (Table 2) because of the heavier representation of the Pennsylvanian poles in the AP population. To correct for this bias, the mean pole for the AP is first moved northwestward along its major axis by 3.2°. This assumes that the angular velocity of the plate is about constant during the 16-m.y. time interval and uses the angular velocity of 0.2°/m.y. determined by *Gordon et al.* [1984] for the Pennsylvanian to Early Triassic time. After this correction, the AP and IC poles still show a significant difference (Figure 7*a*). Possible explanations for this discordance are considered.

One simple explanation is that the systematic discordance between the AP and IC poles was caused by tectonic movement. The difference between the mean poles from Bingham statistics shows that the Appalachians were translated northward by 5.7° ± 3.9° relative to the interior craton, whereas the rotation is not significant (Table 4). If this interpretation is true, then the postulated motion in Pennsylvanian to Late Permian strata is temporally and spatially compatible with the Alleghenian orogenesis in the Appalachians [*Hatcher et al.*, 1989]. North America consists of a stable interior craton to which displaced cratonic margins were attached. After Mississippian time, the whole fold and thrust belt of the Appalachian orogen was welded onto the partially deformed margin of the craton. Geological evidence indicates that the Alleghenian orogeny did not end until Late Permian in the northern, central and southern Appalachians [*Lyones et al.*, 1982; *Hatcher*, 1989], and hence it could have caused relative motion between the Paleozoic AP and IC

TABLE 4. Comparison of the Segments of the Craton

Comparison	Rotation, deg	Translation, deg
AP1‡ versus IC1	-1.7 ± 3.7	5.7 ± 3.7
AP2 versus IC2	-1.7 ± 5.9	8.3 ± 3.1
CP1 versus IC3	5.5 ± 2.9	-2.3 ± 3.4
CP2 versus IC2	8.4 ± 7.4	-2.3 ± 3.9

The error limits are at the 95% confidence level. Positive translation and rotation are northward and clockwise, respectively. For other symbols see the explanation in Table 3.

TABLE 3. Bingham Statistical Analysis of Poles

Region	N*	E_{max}/N_{max}	α_{13}	α_{12}	κ_1	κ_2
AP1	13	126.9/39.9	2.2	3.5	-164.4	-62.5
AP1‡	13	125.0/42.5	2.2	3.5	-164.4	-62.5
IC1	10	124.7/48.2	1.3	4.5	-555.3	-49.8
AP2	4	095.3/49.9	2.2	4.6	-497.3	-118.9
IC2	3	088.1/58.2	1.9	5.4	-953.0	-111.9
CP1	14	112.8/50.3	1.4	5.5	-371.7	-24.6
IC3	15	120.6/48.2	2.1	3.8	-157.7	-45.7
CP2	3	072.2/59.7	2.4	7.4	-588.6	-60.8

E_{max} and N_{max} are longitude (E) and latitude (N) of the mean pole corresponding to the maximum eigenvector using *Bingham* [1974] statistics. α_{13} and α_{12} are approximations of the half-angles at the 95% confidence level directed from the maximum eigenvector toward the minimum and intermediate eigenvectors, respectively. κ_1 and κ_2 are the maximum likelihood estimates of the Bingham concentration parameters. AP1 and AP2 are the AP poles from the Pennsylvanian to Permian and the Middle to Upper Triassic, respectively; IC1 and IC2 are the IC poles from the Pennsylvanian to Permian and the Middle to Upper Triassic, respectively; CP1 and CP2 are the CP poles from the Pennsylvanian to Early Triassic and the Middle to Upper Triassic, respectively; and IC3 is the IC pole from the Pennsylvanian to Early Triassic.

*The number of the poles.

‡The AP1 pole after a 3.2° correction along the trend of the path (see text).

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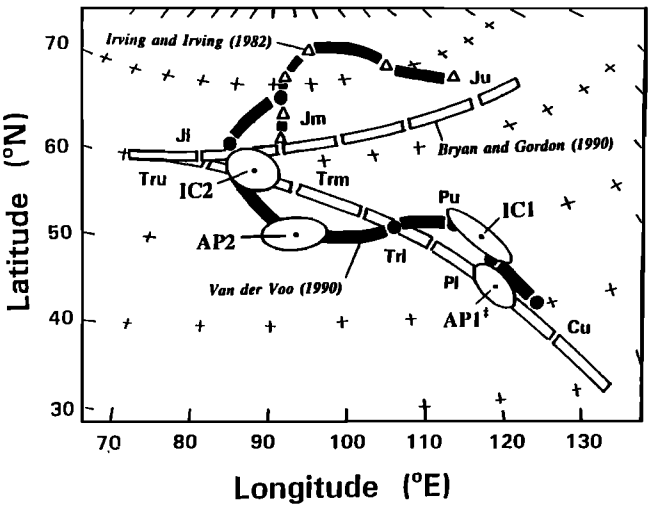


Fig. 6. APW path for North America from *Irving and Irving* [1982], *Bryan and Gordon* [1990], and *Van der Voo* [1990], and the mean pole positions from the AP and IC. Note the north-northwestward shift from the AP to the IC mean poles. The symbols are the same as in Figure 7*a*.

poles. The hypothesis of relative northward displacement of the AP segment is supported by some geological and geophysical evidence for overthrusting of the Appalachians to the northwest [*Cook et al.*, 1979; *Harris and Bayer*, 1979]. A significant problem with this explanation is that the Middle to Upper Triassic data show a similar discordance between the AP and IC poles to that of the Pennsylvanian to Permian data (Figure 7*a*). This implies that the cause of the discordance postdates Triassic time; however, there is no obvious orogenic event or geologic structures to support a post-Triassic translation of the AP segment of the craton relative to the IC segment. Therefore, the tectonic explanation does not appear to be valid.

If the systematic pole discordance in the rock units from the IC and AP segments is not the result of geotectonic motion or an age difference in the rock units, then the other possible cause is that there is a time difference in their magnetization ages. The pole distribution pattern in Figure 6 shows that the mean positions for both the Pennsylvanian-Early Triassic and Middle-Upper Triassic poles from the IC are biased northwestward when compared to the corresponding AP pole means or towards younger portions of the APW path for North America. This pattern could be caused if the IC poles overall have been completely remagnetized or contaminated with an unremoved partial magnetization overprint of slightly younger age, whereas the AP poles have not been completely remagnetized or have been contaminated with a much smaller younger overprint. Therefore, is it possible that the rocks from the IC have been partially or totally remagnetized on average?

In the past decade, the Kiaman remagnetization event has been recognized and documented in many Paleozoic rocks of North America [*Roy and Morris*, 1983; *Irving and Strong*, 1984, 1985; *Kent and Opdyke*, 1978, 1985; *McCabe and Elmore*, 1989]. However, a Kiaman remagnetization overprint cannot explain the systematic pole discordance found in the Triassic rock units, and it also seems impossible that a later heretofore unrecognized pervasive remagnetization event could cause such discordance. A pole discordance is not unknown between rock units from the AP and IC that are deemed to be coeval. Recently, researchers [*Van Fossen and Kent*, 1990; *Butler et al.*, 1992; *Bazard and Butler*, 1992] have found discordant poles in Jurassic rock units from the AP and IC, indicating that some poles were contaminated or remagnetized. More recently, *Witte and Kent* [1990] documented discordance between the Triassic Chinle Formation poles of the IC and the coeval Newark Supergroup poles of the AP. They suggested that the discrepancy could be caused by the Chinle being magnetized over a long time or its primary remanence being contaminated by a younger magnetization overprint.

How the IC rock units were contaminated with a younger remagnetization whereas the AP units were not is still unknown. One possibility is the lithologic difference between the AP and IC rock units. Nearly all of the paleomagnetic data from the IC are from red beds. Conversely, the majority of the data from the AP comes from rock units in which the characteristic magnetization is a

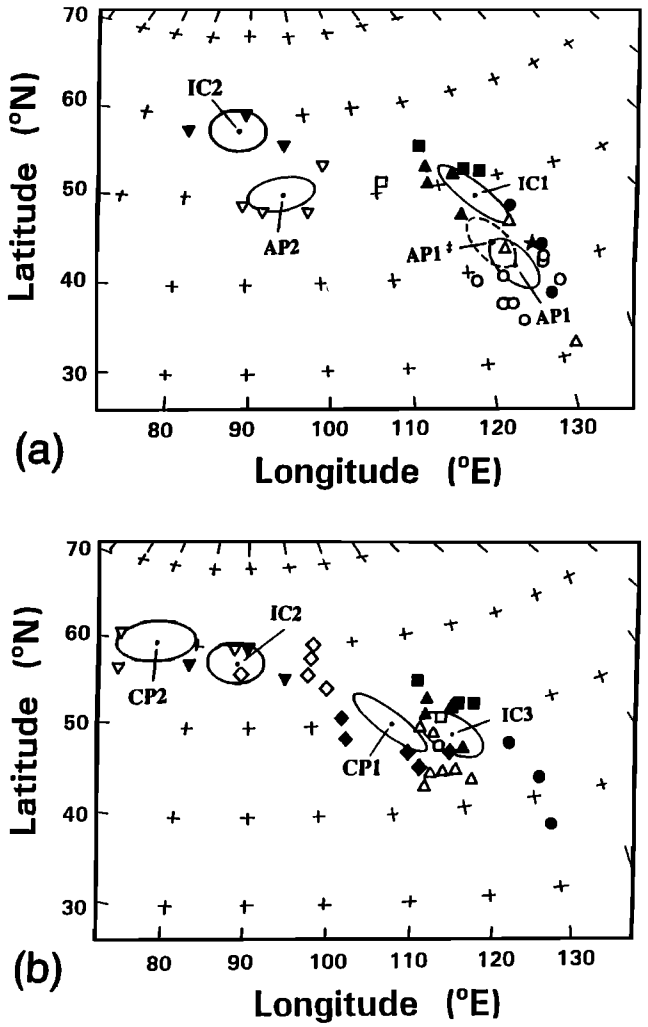


Fig. 7. Pole positions and the mean poles corresponding to the eigenvector corresponding to the maximum eigenvalues with the 95% confidence ovals using *Bingham* [1974] statistics. Solid symbols are Interior craton (IC) poles; circles are Pennsylvanian; triangles are Lower Permian; squares are Upper Permian; diamonds are Lower Triassic; and inverted triangles are Middle-Upper Triassic. (*a*) Comparison of the interior craton (IC) with the Appalachian orogen (AP) segments of the craton. Open symbols are AP poles; star is pole from this study; AP1 and IC1 are the mean poles for the Pennsylvanian to Permian poles; (AP1‡ is the mean pole AP1 after a 3.2° correction along the trend of the path see text) and AP2 and IC2 are the mean poles for the Middle-Upper Triassic poles from the AP and IC, respectively; and (*b*) comparison of the interior craton (IC) with the Colorado Plateau (CP). Open symbols are CP poles; CP1 and IC3 are the mean poles for the Pennsylvanian to Early Triassic poles; and CP2 and IC2 are the mean poles for the Middle to Upper Triassic poles from the CP and IC, respectively.

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primary thermal remanent magnetization (TRM) in magnetite. Paleogeographic and paleoclimatic evidence for the Lower and Middle Jurassic indicates that the formation of secondary chemical remanent magnetization (CRM) in the IC red beds is possible. After the Triassic the western interior of North American was inundated by four transgressions of the sea over a relatively shallow shelf [*Hallam*, 1984]. Paleoclimatic and paleomagnetic reconstructions place the region at a subtropical 25°N latitude [*Smith and Briden*, 1977; *Scotese and Sager*, 1988]. This led to the deposition of aeolian sediments, red beds, evaporites and occasional carbonates with alternating terrestrial, sabka, and shallow marine sedimentation [*Hallam*, 1984]. It also implies a fluctuating water table in the underlying porous late Paleozoic and Triassic clastics through which oxidizing iron-rich solutions would travel. These would be relatively ideal conditions to generate a secondary CRM in these units after their deposition. As Roy and Morris [1983] pointed out, the total remanence of red beds is the sum of the primary detrital remanent magnetization (DRM) carried by specular hematite and magnetite and subsequent secondary CRM. CRM is the common magnetization found in the most red beds. It is carried by hematite from the oxidation of the primary magnetite upon deposition, diagenesis, cementation, metamorphism, or even weathering long after the deposition of the red beds [*Roy and Anderson*, 1981; *Walker et al.*, 1981; *Larson et al.*, 1982]. Secondary CRM can have an unblocking temperature spectrum comparable to, overlapping, or higher than that of any primary CRM or residual DRM. First, it is possible, although unlikely, that the younger magnetization ages in the IC red beds are the result of complete remagnetization at a younger age. Second, although many poles from the IC red beds possess dual polarities that pass a reversal test, this test only indicates that the isolated magnetization is free from much younger magnetization overprints. The reversal test works well when the primary (P) and remagnetization (S) components have distinct unblocking temperature or coercivity spectra and when their directions are more than about 30° apart. It does not work well otherwise.

Consider the case where a red bed acquires its hematite magnetization in two equal pulses some 30 m.y. apart from oxidizing groundwater. P and S would have similar unblocking temperature spectra and be about 15° divergent assuming normal APW. When P and S have the same polarity, either normal (N) or reversed (R), their resultant characteristic remanence (C_N, C_R) will be antiparallel (Figure 8*a* and 8*b*) and represent an average of the P and S directions. Both C_N and C_R will be easily measured and give a pole position that will appear to be primary. When P and S have opposing polarities, they will still give antiparallel remanence (Cn, Cr) directions (Figure 8*c* and 8*d*). However, both Cn and Cr will have significantly weaker intensities and give aberrant pole positions, likely leading to the exclusion of their data upon analysis because of instrumental noise level concerns or, more commonly, statistically aberrant directions. Obviously, this example is a somewhat "special" case, particularly in the assumption that P and S have equal intensities. Nevertheless, the generalization is true that when P and S have similar directions and polarities, then the resultant is enhanced to provide an easily defined "concordant" characteristic remanence vector whereas when they are about antiparallel then the vector is quickly degraded to become poorly defined and discordant. The same basic argument holds if the S remanence is acquired during multiple polarity changes except that the intensity ratio of S_N to S_R becomes important. The net effect of most paleomagnetic data selection methods, whether at the specimen or site level, will be to accept C_N and C_R data but not Cn and Cr data. For this reason, a secondary magnetization may well be present in the IC red beds despite the presence of a positive reversal test. The same may be true for other circumstances such as modern VRM remagnetization in normally magnetized Jurassic rock units [e.g., *Witte and Kent*, 1990].

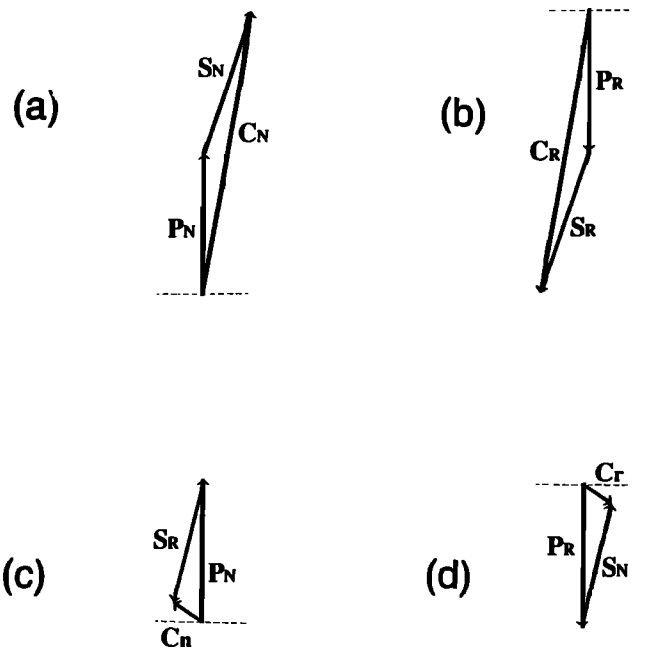


Fig. 8. Schematic showing various relationships between the primary component P and secondary remagnetization component S: (*a*) when S is normal and P is normal, the resultant component is C_N; (*b*) when S is reversed and P is reversed, the resultant component is C_R; (*c*) when S is reversed and P is normal, the resultant component is Cn; and (*d*) when S is normal and P is reversed, the resultant component is Cr.

There is a third possibility that *Van der Voo* [1992] notes. It is that the APW path is not sufficiently well defined in the Pennsylvanian to Triassic time span to permit the determination of such small differences. He suggests that the differences reflect an inadequate data base as well as a variety of minor tectonic rotations and undeciphered remagnetizations. This may be true; nevertheless, the fact that the IC and AP pole populations are statistically different begs a more definitive explanation.

Colorado Plateau Versus Interior Craton

As noted earlier, the Pennsylvanian to Triassic APW path for interior North America consists of two subtracks, one from Pennsylvanian to Early Triassic and the other from Middle to Upper Triassic, for both the CP and IC. Using Bingham statistics it is clear that both are significantly different (Table 4 and Figure 7*b*). They show that the CP has been rotated relative to the IC by 5.5° ± 2.9° and 8.4° ± 7.4° for the Pennsylvanian-Early Triassic and the Middle-Upper Triassic comparisons, respectively, without a significant translation (Figure 7*b*). It is noteworthy that both comparisons show the same sense and comparable magnitudes of rotation. The reason for the larger 8.4° ± 7.4° value is probably that it is not as well constrained because of the small number of the poles available. The rotation of 5.5° ± 2.9° is likely more reasonable because this value fits better with the geologic analysis of *Hamilton* [1981, 1988], who suggested a 2° - 4° clockwise rotation of the Colorado Plateau relative to the craton. This result differs from the analysis of *Steiner* [1988] but is similar to that of *Bryan and Gordon* [1990], who have suggested clockwise rotations of 11° ± 4° and 5.0° ± 2.4°, respectively.

Paleomagnetic poles are statistical values. It is inevitable that every pole will include bias errors from geologic factors such as

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inaccurate age determination, unrecognized local tectonic rotations, or inadequate demagnetization of the samples that are not incorporated into an individual pole's cone of confidence. For this reason the simple comparison of pairs of poles that are deemed to be contemporaneous as done by *Steiner* [1986, 1988] is not an appropriate way to detect rotations of only a few degrees.

Although *Bryan and Gordon*'s [1990] estimate of a 5.0° ± 2.4° clockwise rotation for the CP is very similar to our of 5.5° ± 2.9° estimate, their result indicates a combination of both northward translation and clockwise rotation of the CP relative to the IC. This is because they used a reference point in their analysis that is outside of the Colorado Plateau, whereas our reference point is inside the Colorado Plateau. Further, *Bryan and Gordon* [1986, 1990] used a more selective data base than we have to obtain their paleomagnetic Euler pole (PEP) model solution, and their result is strongly dependent on assumptions built into the model. The PEP model solution, for example, strongly depends on the accuracy of the age assigned to each pole. Furthermore, as *Bryan and Gordon* [1986] noted, the best fit for the rotation of the Colorado Plateau from the PEP model is insensitive to the position of the plateau-craton Euler pole. This means that different combinations of rotation and translation give a similar "best fit" for the PEP model which is not rational in geologic terms. For this reason the PEP method, even though it gives similar results in this case, provides only an approximation so that its use for measuring small tectonic motions is questionable.

CONCLUSION

The Pennsylvanian red beds of the Pictou Group retain a stable remanence in fine-grained hematite which gives a pole position at 42°N, 131°E (dp=2°, dm=4°). Positive conglomerate and contact tests along with dual polarities indicate that the remanence is primary and of Stephanian age. Bingham statistical analysis is more appropriate than Fisher statistics for analyzing groups of poles through time. Such analysis shows that poles from the AP and IC segments of the North American craton from the Pennsylvanian to Late Triassic are significantly different. This discordance cannot be rationalized within the known tectonic framework of the Appalachians nor by a systematic age difference in the rock units from the IC and AP. The likeliest explanation is that this discordance is the result of a younger partial magnetization or possibly complete remagnetization in many of the IC red beds, whereas such overprinting is either much smaller or negligible in the AP rock units. Bingham analysis also shows that there is a significant difference between the IC and CP populations of poles from both the Pennsylvanian-Early Triassic and the Middle-Late Triassic time intervals. The comparison of these poles indicates that there was a 5.5° ± 2.9° clockwise rotation of the Colorado Plateau relative the rest of the craton without a significant northward translation.

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